

MINI PROJECT

Student Performance Analytics using Python, Power BI and Tableau

AIM:

To analyze student academic performance, study habits, learning styles, stress levels and related educational factors using **Python, Power BI and Tableau**, and to create meaningful visualizations that help understand student performance patterns.

DOMAIN:

Data Analytics and Visualization Sub-Domain:

Educational / Student Performance Analytics

PROBLEM STATEMENT:

Student academic performance can vary based on study hours, attendance, assignment completion, learning style, stress level, online courses and other educational factors.

The project focuses on analyzing student data to identify:

- Study hours and exam performance
- Attendance patterns
- Assignment completion
- Learning style distribution
- Stress level distribution
- Exam score variations
- Final grade distribution
- Online course participation
- Educational technology usage
- Relationship between study hours and exam scores

A data analytics and visualization system is therefore required to convert the raw student data into understandable charts and dashboards.

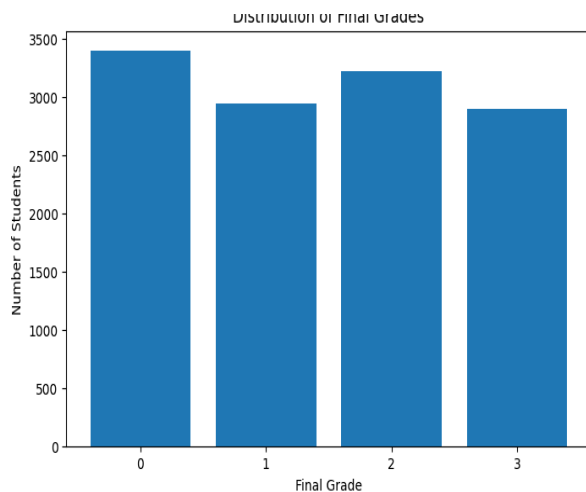
OBJECTIVES:

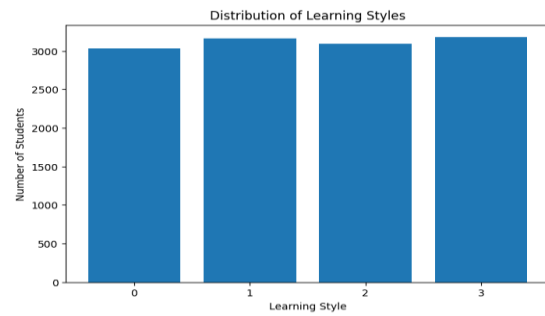
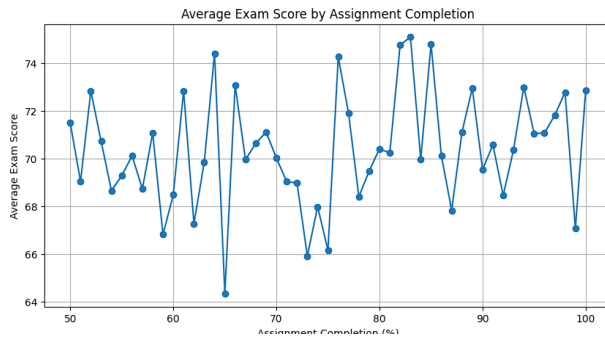
- To collect and organize student performance data.
- To clean and prepare the dataset for analysis.
- To analyze student academic performance.
- To study the distribution of final grades.
- To analyze learning styles and stress levels.
- To examine the relationship between study hours and exam scores.
- To visualize the results using Python.
- To create an interactive dashboard using Power BI.
- To create visual analytics using Tableau.
- To derive useful insights from the visualizations.

OUTPUT:

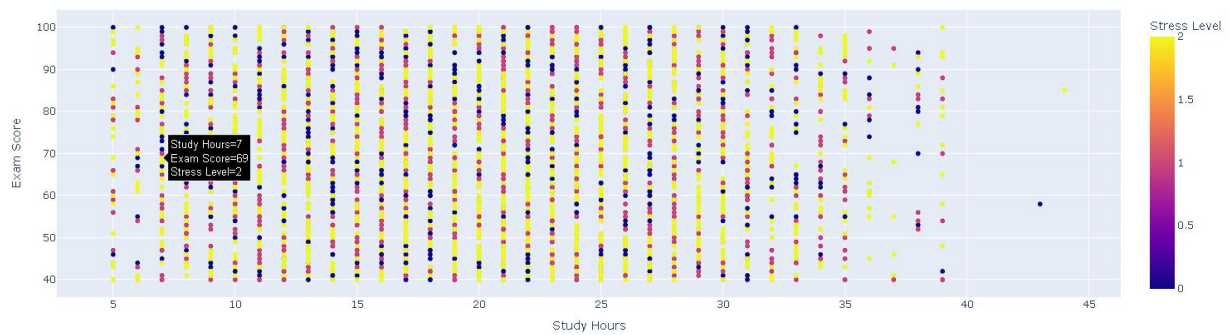
Python :

	StudyHours	Attendance	Resources	Extracurricular	Motivation	Internet	Gender	Age	LearningStyle	OnlineCourses	Discussions	AssignmentCompletion	ExamScore	EduTech	StressLevel	FinalGrade
0	19	64	1	0	0	1	0	19	2	8	1	59	40	0	1	3
1	19	64	1	0	0	1	0	23	3	16	0	90	66	0	1	2
2	19	64	1	0	0	1	0	28	1	19	0	67	99	1	1	0
3	19	64	1	1	0	1	0	19	2	8	1	59	40	0	1	3
4	19	64	1	1	0	1	0	23	3	16	0	90	66	0	1	2





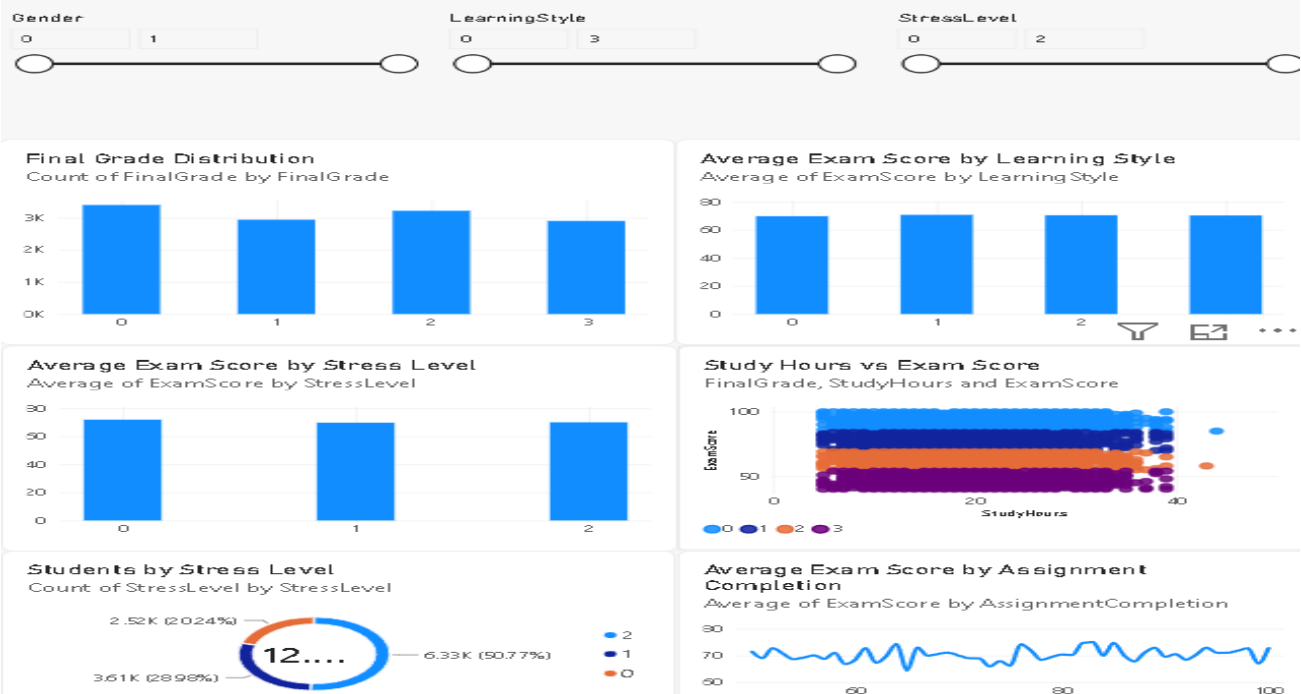
Study Hours vs Exam Score by Stress Level



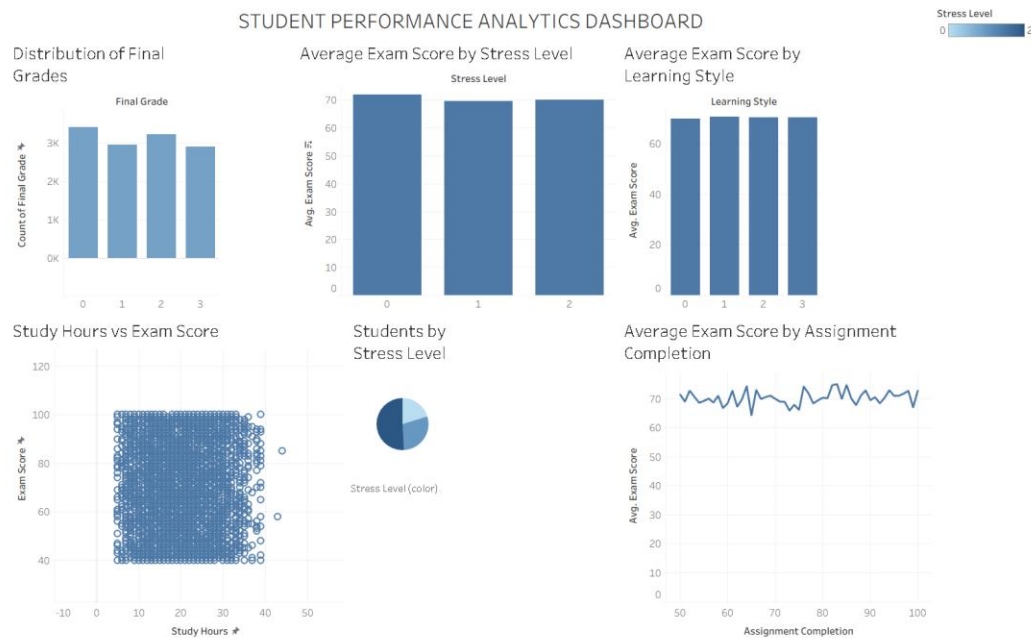
Study Hours: 5 to 44
Attendance: 60 to 100
Age: 18 to 29
Online Courses: 0 to 20
Assignment Completion: 50 to 100
Exam Score: 40 to 100

POWER BI:

STUDENT PERFORMANCE ANALYTICS DASHBOARD



TABLEU:



RESULT:

The **Student Performance Analytics** project was implemented using **Python, Power BI and Tableau**.

Python was used for data loading, cleaning, preprocessing, analysis and visualization. Power BI was used to create an interactive dashboard containing key analytical charts and filters.

Tableau was used to create interactive visualizations for exploring student performance, learning styles, stress levels and academic patterns.

The project demonstrates how data analytics and visualization techniques can be applied to student data to understand **academic performance, study patterns, learning styles and related educational factors**.